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S.No. (Poster Code):

Fair and Quality-Aware Diffusion through Dual Optimization

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Abstract

Diffusion models are powerful generative tools, but often replicate biases present in training data, raising concerns about fairness. The Constrained Diffusion Model via Dual Training addresses this by framing generation as a constrained optimization problem, enforcing fairness through distributional constraints. In this work, we build upon that framework and introduce an improved constraint formulation aimed not only at fair sampling, but also at improving sample quality. Our approach incorporates perceptual and diversity-based regularizers into the dual optimization scheme, encouraging the model to generate samples that are both equitable and high fidelity. Through empirical evaluation on benchmark datasets, we demonstrate that this extended formulation achieves a better trade-off between fairness and quality.

Objective

The primary objective of this research is to design a principled and generalized **constrained diffusion framework** that:

- Promotes **fairness** by ensuring the model respects underrepresented data distributions,
- Enhances the **quality of generated samples**,
- Accommodates **multiple generation requirements** within a unified framework.

This is achieved by formulating the training of diffusion models as a *constrained distribution optimization problem*, where the main loss is minimized with respect to the original data distribution, while auxiliary constraints enforce low loss with respect to other desirable distributions.

Methodology

Introducing a parametrized model p_θ, the constrained optimization problem via a score function $\widehat{s}_\theta$ is given by:

$$\min_\theta E_q\|\widehat{s}_\theta - \nabla \log q\|^2 \quad \text{s.t.} \quad E_{q^i}\|\widehat{s}_\theta - \nabla \log q^i\|^2 \le \tilde{b}^i \qquad \text{(P-LOSS)}$$

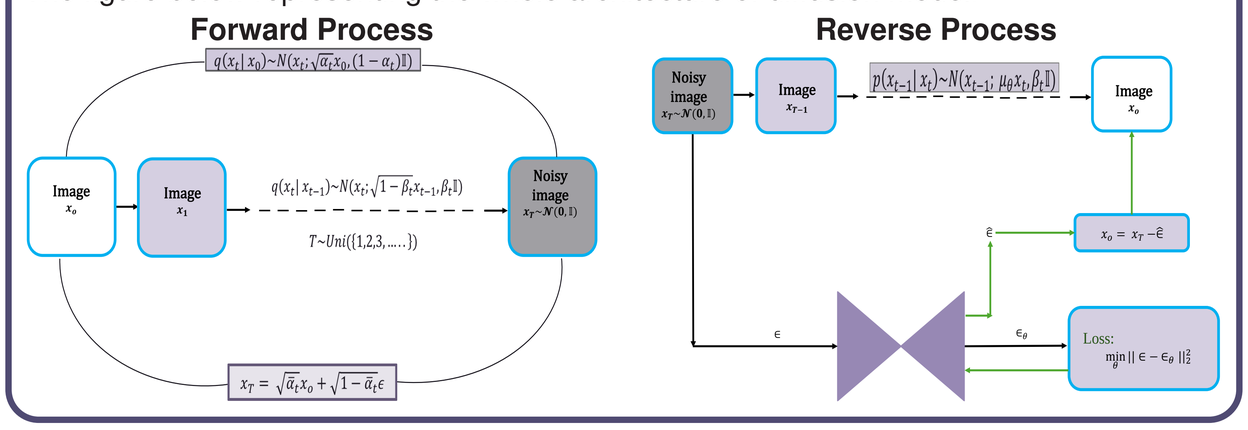
Score-Based Lagrangian: Define the Lagrangian for the score-based problem as:

$$\overline{\mathcal{L}}_s(\theta,\lambda) = E_q\|\widehat{s}_\theta - \nabla \log q\|^2 + \sum_{i=1}^{m} \lambda_i \left(E_{q^i}\|\widehat{s}_\theta - \nabla \log q^i\|^2 - \tilde{b}^i\right)$$

Dual Formulation (Score-Based): The dual of the score-based constrained problem is given by:

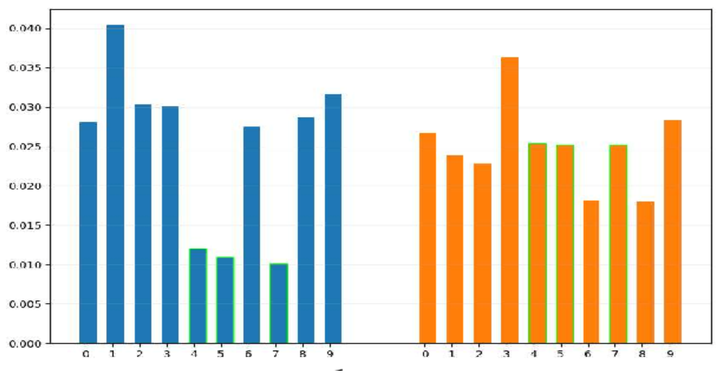
$$\bar{g}_s(\lambda) = \min_\theta \overline{\mathcal{L}}_s(\theta,\lambda), \quad \bar{\lambda}_s^\star = \arg\max_{\lambda \ge 0} \bar{g}_s(\lambda)$$

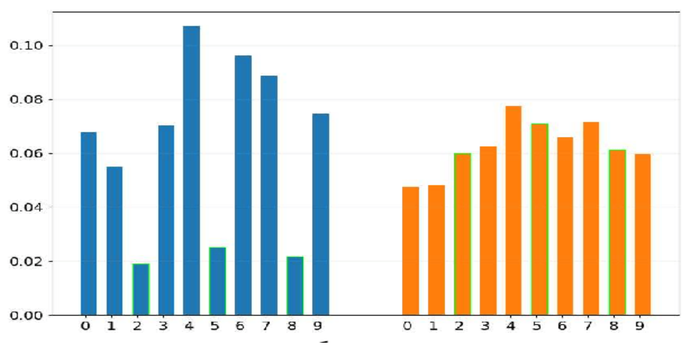
The figure below representing the whole architecture of diffusion model.

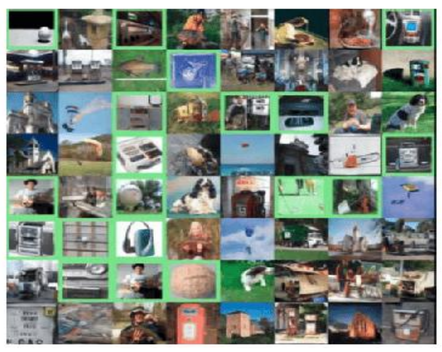


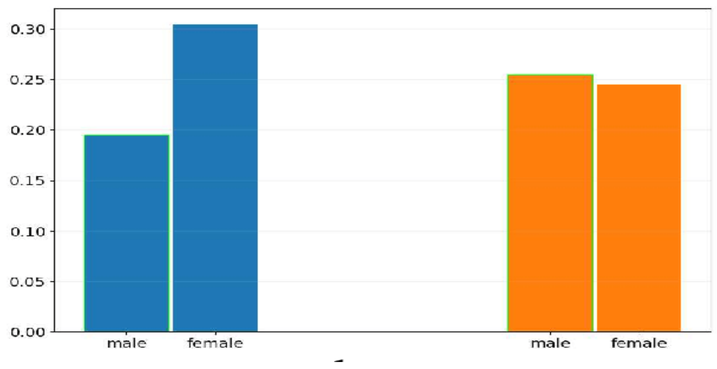
Experiments and Results

Minority Class (MC) Experiment: We evaluate our approach on MNIST, Celeb-A, and ImageNet. To simulate minority representation, we first construct a balanced dataset for q by sampling equally from each class and then remove samples from specific classes to make them underrepresented. Each q^i corresponds to one such underrepresented class. Our constrained model (●) generates samples more uniformly across classes, especially improving generation for underrepresented ones, compared to the unconstrained model (●).









Datasets	MNIST	Celeb-A	ImageNet
FID ●	15.9	19.6	36.0
FID ●	13.4	11.6	27.3

Datasets	Minority classes
MNIST	4,5,7
ImageNet	Cassette player, French horn, Golf ball
Celeb-A	Male

Parameters of U-net Model used as noise predictor	
# ResNet layers per U-Net block	2
# down/upsampling blocks	6
# Output channels for U-Net blocks	(128, 128, 256, 256, 512, 512)

Conclusion

Our method effectively reduces bias on 3 benchmark datasets by generating samples from an optimal mixture distribution by imposing constrained for fair and quality sampling, without relying on pretraining or data reweighting. This framework is general, extensible, and lays the groundwork for future research on domain-specific constraints, conditional generation, and improved convergence in diffusion models.

References

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